

Course Code: B20 BS 2102								
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20			
II B. Tech I Semester - MODEL QUESTION PAPER								
NUMERICAL METHODS AND ADVANCED CALCULUS								
(Common to ECE & ME)								
TIME: 3 Hrs.			Max. Marks: 70 M					
Answer any One Question from Each Unit								
All questions carry equal marks								
	UNIT-I					CO	KL	M
1.a)	Find the root of the equation $2x - \log_{10} x = 7$, using Newton -Raphson method.					1	3	7
b)	Find the root of the equation $x^3 - x - 1 = 0$, using Regula - Falsi method.					1	3	7
OR								
2. a)	Using Newton's Forward Interpolation formula find $f(5)$ for the following data					1	3	7
	x	4	6	8	10			
	y= f(x)	1	3	8	16			
b)	Using Lagrange's interpolation formula to find value of y when x = 10 from the following					1	3	7
	x	5	6	9	11			
	y	12	13	14	16			
UNIT-II								
3.a)	Use the Trapezoidal rule to estimate the $\int_0^2 e^{x^2} dx$, by taking 8 intervals.					2	3	7
b)	Solve by Taylor's series method the equation $\frac{dy}{dx} = \log xy$, for $y(1.1)$ given that $y(1) = 2$.					2	3	7
OR								
4. a)	Find an approximate value of y when $x = 0.1$, if $\frac{dy}{dx} = x - y^2$ and $y = 1$ at $x = 0$, using Picard's method.					2	2	7
b)	Use fourth order Runge-Kutta method to find y at $x = 0.1$, given that $\frac{dy}{dx} = 3e^x + 2y$, $y(0) = 0$ and $h = 0.1$.					2	3	7
UNIT-III								
5.a)	Evaluate $\iint xy \, dx \, dy$ over the 1st quadrant of the circle $x^2 + y^2 = a^2$.					3	3	7
b)	Evaluate by changing the order of integration $\int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx$.					3	3	7
OR								
6. a)	Evaluate $\int_0^1 \int_0^{\sqrt{1-y^2}} (x^2 + y^2) \, dx \, dy$ by changing into polar coordinates					4	3	7
b)	Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz \, dx \, dy \, dz$					4	3	7
UNIT-IV								
7. a)	Find a unit vector normal to the surface $xy^3z^2 = 4$ at the point $(-1,-1,2)$					5	3	7

b)	Find the directional derivative of $f(x, y, z) = xy^3 + yz^3$ at the point (2,-1,1) in the direction of vector $\bar{I} + 2\bar{J} + 2\bar{K}$.	5	3	7
OR				
8.	Verify Green's theorem for $\oint_C [(3x - 8y^2)dx + (4y - 6xy)dy]$ where 'C' is the boundary of the region bound by $x=0$, $y=0$ and $x+y=1$.	5	3	7
UNIT-V				
9.a)	Prove that the function defined by $f(z) = \frac{(1+i)x^3 - (1-i)y^3}{x^2 + y^2}$ ($z = 0$), and $f(0)$ is continuous and the Cauchy Riemann equations are satisfied at the origin, yet $f'(0)$ do not exist.	6	3	7
b)	Find the complex potential w and velocity potential ϕ when its stream function is $\psi = \tan^{-1}\left(\frac{y}{x}\right)$	6	3	7
OR				
10.a)	If $f(z)$ is a regular function of z , prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) f(z) ^2 = 4 f'(z) ^2$	6	3	7
b)	Find the bilinear transformation that maps the points, $(\infty, i, 0)$, into the point $(0, 1, \infty)$.	6	3	7

Course Code:B20EC2101						
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R 20	
II/IV B.Tech I Semester (R20) Regular Examinations						
ELECTRONIC DEVICES AND CIRCUITS						
Common to ECE& EEE						
Time: 3 Hrs.			Max. Marks:70			
Answer any one Question from Each Unit						
All questions carry equal Marks						
				CO	KL	M
UNIT-I						
1	a	Discuss in detail the various current components in a p-n junction diode.	1	3	7	
	b	A Silicon diode operates at a forward voltage of 0.4 V. Calculate the factor by which the current will be multiplied when the temperature is increased from 25 degree centigrade to 150 degrees centigrade.	1	3	7	
OR						
2	a	With a neat diagram, explain the operation of a full wave rectifier and obtain expressions for ripple factor and efficiency.	1	3	7	
	b	Distinguish between Zener breakdown and Avalanche breakdown.	1	2	7	
UNIT-II						
3	a	Explain input and output characteristics of the transistor in CE configuration with a neat sketch.	1	2	7	
	b	Compare CE, CC and CB configurations of a transistor.	1	3	7	
OR						
4	a	Explain the early effect and its consequences in the Common base transistor.	1	2	7	
	b	Define α , β , γ and write the relation between them	1	2	7	
UNIT-III						
5	a	Explain how a self-bias circuit improves stability of the operating point and obtain an expression for the stability factor 'S' for the self-bias circuit.	2	2	7	
	b	Explain how diodes can be used to compensate against variations in V_{BE} and I_{CO} due to change in temperature.	2	3	7	
OR						
6	a	A transistor with $\beta= 100$ is to be used in Common Emitter Configuration with collector to base bias. $R_E = 1\text{ K}\Omega$, $V_{CC} = 10\text{V}$. Assume $V_{BE} = 0.7\text{V}$. Choose R_B so that the quiescent collector to emitter voltage is 4V. Find Stability Factor S.	2	3	7	
	b	Explain thermal runaway and thermal stability in transistors.	2	2	7	
UNIT-IV						
7	a	Compare JFET and BJT.	3	3	7	

	b	Draw the basic structure of N- channel JFET and explain the operation with the help of characteristic curves.	3	3	7
OR					
8	a	Explain the construction and operation of Enhancement MOSFET with drain and transfer characteristics.	3	2	7
	b	Derive the expression for pinch off voltage of JFET.	3	4	7
UNIT-V					
9	a	Derive expressions for current gain, Voltage gain, Input impedance and Output impedance of a generalized transistor amplifier using h-parameters.	4	4	7
	b	Explain the frequency response of the CE amplifier with a neat sketch.	4	2	7
OR					
10	a	Discuss the effect of bypass and coupling capacitors in CE amplifiers.	4	3	7
	b	Draw and explain a hybrid - model of a transistor used at high frequencies.	4	4	7

UNIT-IV																																																													
7	a.	Convert D-FF into JK-FF.	3	3	7																																																								
	b.	What is a race around problem in a JK flip-flop and explain the method to avoid it by using Master Slave JK-FF.	3	4	7																																																								
OR																																																													
8	a.	Draw a 3-bit right shift register and explain the modes SIPO, SISO, PIPO and PISO.	3	4	7																																																								
	b.	Design Mod-5 Ring counter and draw its output waveforms.	3	3	7																																																								
UNIT-V																																																													
9	a	Explain the difference between synchronous and asynchronous circuits.	4	3	7																																																								
	b	Design a Sequence detector which produces an output 0 every time the sequence 0101 is detected, an output 1 at all other times. Assume that an overlapping sequence is allowed. Use D-FFs in your design.	4	4	7																																																								
OR																																																													
10	a.	Design a synchronous sequential circuit using D-FFs for the given state table. <table><tr><th colspan="2">Present State</th><th colspan="4">Next State</th><th colspan="2">Output</th></tr><tr><th colspan="2"></th><th colspan="2">X=0</th><th colspan="2">X=1</th><th>X=0</th><th>X=1</th></tr><tr><th>A</th><th>B</th><th>A</th><th>B</th><th>A</th><th>B</th><th>A</th><th>B</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr></table>	Present State		Next State				Output				X=0		X=1		X=0	X=1	A	B	A	B	A	B	A	B	0	0	0	0	0	1	0	0	0	1	0	0	1	1	1	0	1	0	0	0	1	0	1	0	1	1	0	0	1	0	1	0	4	4	7
Present State		Next State				Output																																																							
		X=0		X=1		X=0	X=1																																																						
A	B	A	B	A	B	A	B																																																						
0	0	0	0	0	1	0	0																																																						
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1	0	0	0	1	0	1	0																																																						
1	1	0	0	1	0	1	0																																																						
	b.	What is lockout in counters? Design a Mod-10 synchronous counter by using D-Flip- Flops.	4	3	7																																																								

II B. Tech I Semester Regular Examinations

SIGNALS AND SYSTEMS

Electronics and Communication Engineering

Time: 3 Hrs.

Max. Marks: 70

Answer any one Question from Each Unit

All questions carry equal Marks

CO KL M

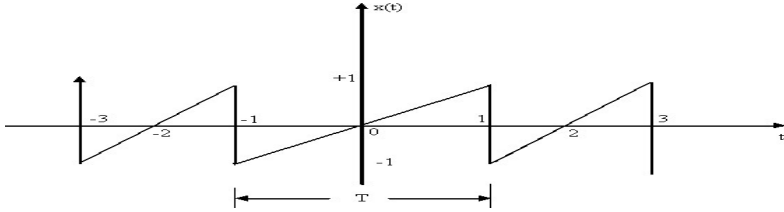
UNIT-I

1	a.	Explain all classification of signals with examples for each category.	1	2	7
	b.	Determine the power and RMS value of the following signals. 1. $x(t) = 5 \cos \cos(50t + 3)$ 2. $x(t) = 10 \cos \cos(5t) \cos(10t)$	1	3	7

OR

2	a.	Determine whether the following systems are time invariant or not. 1. $y(t) = x(t^2)$ 2. $y(n) = x(2n)$	1	3	7
	b.	Prove the energy of the power signal is infinite over infinite time.	1	2	7

UNIT-II

3	a.	Find the trigonometric Fourier series for the periodic signal $x(t)$ as shown in figure below 	2	3	7
	b.	State and prove parseval's relation of Fourier series.	2	3	7

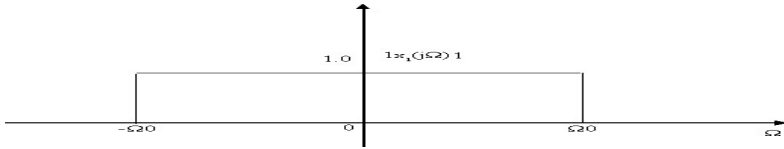
OR

4	a.	State and prove the time shifting and time scaling properties of CTFS.	2	3	7
	b.	Explain Dirichlet conditions on Convergence of Fourier Series.	2	2	7

UNIT-III

5	a.	State and prove the differentiation in the frequency domain property of CTFT.	3	3	7
	b.	Find $x(t)$ from $X(j\omega) = 1/(1+j\omega)^2$	3	3	7

OR

6	a.	Find the inverse Fourier transform of $X(j\Omega)$ for the spectra shown in figure below.	3	3	7
					
	b.	Find the Fourier transform of the following 1. $e^{at}u(-t)$ 2. $te^{-at}u(t)$	3	3	7
UNIT-IV					
7	a.	State and prove the final value theorem of Laplace transform.	4	3	7
	b.	Compute the initial and final values for $x(s) = \frac{2s+5}{s(s+3)(s+4)^2}$	4	2	7
OR					
8	a.	Explain ROC properties of Laplace transforms.	4	2	7
	b.	Determine the bilateral Laplace transform and ROC for the following signals. 1. $e^{-2t}u(t-2)$ 2. $e^{-4 t }$	4	3	7
UNIT-V					
9	a.	Find the inverse z-transform of the following: i) $X(z) = \frac{1}{(1+3z^{-1}+2z^{-2})}$ ROC: $\text{mod}(Z) > 2$ ii) $X(z) = \frac{(Z+1)}{(4+35+Z^2)}$ ROC: $\text{mod}(Z) < 2$	5	3	7
	b.	Solve the following difference equation for $y(n)$ using z-transform for the specified Initial condition. $y(n) - y(n-1) + \frac{1}{4}y(n-2) = X(n) : n > 0$ where $X(n) = 2\left(\frac{1}{8}\right)^n$ $y(-1) = 2$ and $y(-2) = 4$.	5	3	7
OR					
10	a.	A signal $x(t) = \sin c(150\pi t)$ is sampled at a rate of (i) 100Hz (ii) 200 Hz (iii) 300Hz. For each of these cases, explain if you can recover the signal $x(t)$ from the sampled signal.	5	3	7
	b.	Find the z-transform and ROC for the following: 1. $x(n) = a^n u(n)$ 2. $x(n) = -b^n u(-n-1)$	5	3	7

Course Code:B20EC2104						
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R 20		
II B. Tech I Semester Regular Examinations						
RANDOM VARIABLES AND STOCHASTIC PROCESSES						
Electronics and Communication Engineering						
Time: 3 Hrs.				Max. Marks:70		
Answer any one Question from Each Unit						
All questions carry equal Marks						
				CO	KL	M
UNIT-I						
1	a.	State and prove Bayes theorem.	1	3	7	
	b.	A single card is drawn from a 52 card deck (i) What is the probability that the card is a jack? (ii) What is the probability that the card will be a 5 (or) smaller? (iii) What is the probability that the card is a red 10?	1	3	7	
OR						
2	a.	State and prove the Addition theorem of probability.	1	3	7	
	b.	In a classroom, 70% are above average, 20% are average and 10% are below average. Suppose that 20% of above average, 10% of average and 20% of below average students fail in a subject. What is the probability that a randomly selected student is an average student who failed?	1	3	7	
UNIT-II						
3	a.	Define Distribution and Density functions. List the properties of them. Prove any one property of each.	2	2	7	
	b.	For real constants $b > 0, c > 0$, and any a , find a condition on constant a and a relationship between c and a (for a given b) such that the function $f_X(x) = a[1 - (x / b)] 0 \leq x \leq c$ $= 0 \quad \text{elsewhere}$ Is a valid probability density.	2	3	7	
OR						
4	a.	The joint probability density function of two random variables X and Y is given by $f_{XY}(x,y) = k e^{-(ax+by)} \text{ if } x \geq 0, y \geq 0 ,$ $= 0 \text{ otherwise}$ Find the value of k . Also prove that X and Y are independent.	2	3	7	
	b.	A random variable X has a Gaussian density function with mean '0'and variance '1'. Find the probability density function of the random variable Y defined as $Y = X^2$.	2	3	7	
UNIT-III						
5	a.	State and prove Chebyshev's inequality .	3	3	7	
	b.	Consider that a pdf of a random variable X is $f_X(x) = 1/ K-2 \leq x \leq 3$ and another random variable $Y = 2X$ $= 0 \text{ otherwise}$ Then find value of $K, E[X], E[Y]$ and $E[XY]$.	3	3	7	

OR					
6	a.	Find the characteristic function for a random variable x with density function $f_x(x) = e^{- x }$ $x \geq 0$	3	3	7
	b.	State and Prove Central Limit Theorem.	3	3	7
UNIT-IV					
7	a.	Define Autocorrelation and power spectral density for a random process. List their Properties.	4	2	7
	b.	Determine the PSD of a stationary process whose autocorrelation is given by $R_{xx}(\tau) = \sigma^2 e^{-\alpha \tau }$	4	3	7
OR					
8	a.	Prove that the random process is wide sense stationary $X(t) = A \cos(\omega_0 t + \theta)$ where A, ω_0 are constants and θ is a uniformly distributed random variable on the interval $(0, 2\pi)$.	4	3	7
	b.	Distinguish between stationary and Ergodic random processes.	4	2	7
UNIT-V					
9		Find the input autocorrelation function, output autocorrelation function and output spectral density of RC low pass filter, when the filter is subjected to a white noise of spectral density $N_0/2$.	5	3	14
OR					
10	a.	Show that the Narrow band noise process can be expressed as In-phase and Quadrature components.	5	3	7
	b.	Write short notes on (i) Poisson Process (ii) Gaussian Process	5	2	7

II B. Tech II Semester Regular Examinations

ELECTRONIC CIRCUIT ANALYSIS AND DESIGN

(Common to ECE & EEE)

Time: 3 Hrs.

Max. Marks: 70

Answer any one Question from Each Unit

All questions carry equal Marks

			CO	KL	M
UNIT-I					
1	a.	Derive the expressions for gain of an BJT RC coupled amplifier at low frequency.	1	3	7
	b.	A transistor is connected as a CE amplifier with load resistance of $10\text{K}\Omega$. The parameters are $h_{ie}=5\text{K}\Omega$ and $h_{fe}=330$. Calculate the overall gain for mid frequency range when four such stages are connected in cascaded RC coupling. Assume $R_s=0$.	1	3	7
OR					
2	a.	Explain what you understand by the term bandwidth shrinkage factor as applied to cascaded RC stages. Determine the expression for the same.	1	3	7
	b.	A two stage RC coupled amplifier uses the transistor whose h-parameters are $h_{ie}=4.5\text{K}\Omega$, $h_{fe}=330$. If the load resistance $R_L=5.5\text{K}\Omega$ then find the required value of coupling capacitor C_C , so that the lower cut-off frequency is 60 Hz.	1	2	7
UNIT-II					
3	a.	Determine expressions for input impedance and output impedance in case of a voltage series feedback topology.	2	2	7
	b.	An amplifier gain change by $\pm 10\%$ using negative feedback amplifier is to be modified to yield gain of 100 with 0.1% variation. Calculate the required loop gain and amount of negative feedback.	2	2	7
OR					
4	a.	Determine expressions for input impedance and output impedance of voltage –shunt and current-shunt topologies and identify the types of amplifiers.	2	3	7
	b.	A negative feedback of $\beta=0.002$ is applied to an amplifier of gain 1000. Calculate the change in overall gain of the feedback amplifier, if the internal amplifier is subject to a gain reduction of 15%.	2	3	7
UNIT-III					
5	a.	Explain the working of RC phase shift oscillator using BJT and derive frequency of oscillations.	3	3	7
	b.	In a colpitt's oscillator $C_1=0.02\text{ }\mu\text{F}$ and $C_2=0.2\text{ }\mu\text{F}$. if the frequency of oscillator is 10KHz, Find the value of inductor and also find the required gain for oscillator.	3	3	7
OR					

6	a.	Obtain the frequency and condition for sustained oscillations of colpitt's oscillator	3	3	7
	b.	In Hartley oscillator, calculate L2 if L1=15 mH, C = 50 pF, mutual inductance of 5 μ H and the frequency of oscillations is 168 KHz.	3	3	7
UNIT-IV					
7	a.	Obtain frequency response of staggered tuned amplifier	4	3	7
	b.	Explain the working of transformer coupled class A power amplifier and calculate it's efficiency. Discuss its merits and demerits	4	3	7
OR					
8	a.	Determine frequency response of single tuned amplifier	4	3	7
	b.	Explain the working of class B push pull power amplifier and calculate it's efficiency. Discuss its merits and demerits.	4	3	7
UNIT-V					
9	a.	Analyze dual input unbalanced output differential amplifier	5	4	7
	b.	For a dual input unbalanced output differential amplifier, $V_{CC}=13V, V_{EE}=-13V, R_C=4.7K, R_E=6.8K, R_S=50\Omega$. Determine (i)Q point values (ii) Voltage gain (iii)differential input resistance (iv)output resistance.	5	4	7
OR					
10	a.	Analyze the practical differentiator circuit.	5	4	7
	b.	Evaluate the output voltage of op-amp for input voltages 150 μ V and 140 μ V. If the differential gain is 4000V and value of CMRR is 10^5 .	5	3	7

Course Code:B20EC2202										
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)								R 20		
II B. Tech I Semester Regular Examinations										
LINEAR AND DIGITAL IC APPLICATIONS										
(For ECE)										
Time: 3 Hrs.						Max. Marks:70				
Answer any one Question from Each Unit										
All questions carry equal Marks										
								CO	KL	M
UNIT-I										
1	a.	Draw a block diagram of typical OP-AMP and explain the function of each block.						1	3	7
	b.	Explain the working of Instrumentation amplifier using 3 Op-Amps.						1	3	7
OR										
2	a.	Explain the operation of Square wave generator circuit with neat circuit diagram and derive expression for time period.						1	3	7
	b.	Explain the operation of an Regenerative comparator with circuit diagram and Waveforms.						1	3	7
UNIT-II										
3	a.	Derive the transfer function of a second order LPF. Comment on its frequency response.						2	3	7
	b.	Why active filters are preferred to passive filters. Explain the operation of All-pass filters with neat circuit diagrams and waveforms.						2	2	7
OR										
4	a.	Design a Wide Band Pass Filter with $f_L = 200\text{Hz}$, $f_H = 1\text{kHz}$ and pass band gain of 4 Calculate the value of Q for the filter.						2	3	7
	b.	Draw a block diagram of typical voltage controlled oscillator and explain its working.						2	3	7
UNIT-III										
5	a.	Draw the circuit of Schmitt trigger using IC555 timer and explain its operation?						3	3	7
	b.	Design a monostable multivibrator using 555 timer to produce a pulse width of 100 ms and explain its working with the help of a functional diagram of a 555 timer.						3	3	7
OR										
6	a.	With the help of schematic diagram of 555 timer, explain how it can be used as mono stablemultivibrator						3	3	7
	b.	Draw and explain the working of 555-timer circuit in astable mode to get output waveform with 50% duty cycle						3	3	7

UNIT-IV					
7	a.	With a neat diagram explain the working principle of R-2R ladder type DAC.	4	3	7
	b.	With neat block diagram, explain successive approximation type A/D converter in detail	4	3	7
OR					
8	a.	Which is the fastest ADC? Explain the operation and discuss its merits and de-merits	4	3	7
	b.	Draw and explain the circuit diagram of parallel comparator type ADC.	4	3	7
UNIT-V					
9	a.	Draw the circuit of a TTL NAND gate and explain the operation.	5	3	7
	b.	Write a VHDL program to build half and full-adder circuits.	5	3	7
OR					
10	a.	Describe a basic ECL Nor gate and explain its working with the suitable truth table and diagram?	5	3	7
	b.	Write a VHDL Code for Binary Comparator	5	3	7

	b.	Draw the block diagram of FM transmitter and explain its operation.	4	K2	7
OR					
8	a.	Explain the principle of operation of a super heterodyne radio receiver.	4	K2	7
	b.	Distinguish between AGC and Delayed AGC with help of neat circuit diagrams.	4	K4	7
UNIT-V					
9	a.	Derive the expression for signal -to-noise ratio in DSB-SC system.	5	K3	8
	b.	Explain Threshold in FM.	5	K2	6
OR					
10	a.	Explain the modulation and demodulation techniques for pulse time modulation systems.	5	K2	8
	b.	Explain Time Division Multiplexing.	5	K2	6

Course Code:B20EC2204								
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R 20			
II B. Tech II Semester Regular Examinations								
COMPUTER ARCHITECTURE AND ORGANIZATION								
Electronics and Communication Engineering								
Time: 3 Hrs.			Max. Marks:70					
Answer any one Question from Each Unit								
All questions carry equal Marks								
					CO	KL	M	
UNIT-I								
1	a.	Design and explain 4-bit arithmetic circuit with functional table				1	3	7
	b.	The adder- subtractor circuit has following values for input mode M and data inputs A and B. In each case, determine the values of the outputs: S ₃ ,S ₂ ,S ₁ ,S ₀ and C ₄ .				1	3	7
		M	A	B				
		0	0111	0110				
		0	1000	1001				
		1	1100	1000				
		1	0101	1010				
		1	0000	0001				
OR								
2	a.	Explain the operation of a 4-bit adder-subtractor with a neat diagram.				1	2	7
	b.	Design a common bus system with multiplexers to connect four 4 bit registers A,B,C,D. How this can be modified with the help of a three state buffer.				1	3	7
UNIT-II								
3	a.	Explain the data transfer, Logic and Program Control Instructions				2	2	7
	b.	With the help of a flowchart explain the operation of a basic computer during 1. Instruction cycle 2. Interrupt cycle				2	3	7
OR								
4	a.	With the help of a diagram explain the functions of the basic computer registers.				2	3	7
	b.	Explain in detail about basic instruction formats with examples.				2	2	7
UNIT-III								
5	a.	Explain the concept of microprogramming in detail with the help of an				3	2	7

		exmple			
	b.	Give an overview of address sequencing in microprogrammed control unit.	3	2	7
OR					
6	a.	With an example, explain various addressing modes and write how to calculate the address in each case.	3	2	7
	b.	What is a stack? Discuss its organization.	3	3	7
UNIT-IV					
7	a.	Explain Daisy Chain Interrupt Mechanism. List the advantages and disadvantages in this scheme	4	2	7
	b.	With the help of block diagram and truth table explain the operation of an I/O interface unit`	4	3	7
OR					
8	a.	Show internal configuration of DMA and explain its operation in a computer for read and write modes.	4	3	7
	b.	Give an overview of parallel priority interrupts hardware.	4	3	7
UNIT-V					
9	a.	A system employs RAM chips of 256 bytes and ROM chips of 1024 bytes. It needs 2KB RAM, 4KB ROM and four interface units with four registers each and memory mapped I/O used. The two highest order bits of the address bus are 00 for RAM, 01 for ROM and 10 for interface registers. Draw a memory address map for the system	4	4	7
9	b	Explain Set Associative mapping for organizing cache memory.	4	3	7
OR					
10	a.	Explain Memory hierarchy and Differentiate between Main memory and Auxiliary memories	4	3	7
	b.	Explain the following 1. Associative Memory2. Virtual Memory	4	3	7

Course Code:B20EC2205							
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)						R 20	
II B. Tech II Semester Regular Examinations							
ELECTROMAGNETIC FIELD THEORY AND TRANSMISSION LINES							
Electronics and Communication Engineering							
Time: 3 Hrs.				Max. Marks:70			
Answer any one Question from Each Unit							
All questions carry equal Marks							
				CO	KL	M	
UNIT-I							
1	a.	A charge $Q=-10\text{nc}$ is at the origin in free space. If the x-component of $\mathbf{E}$ is to be zero at the point (3,1,1), determine the value Q_t which should be kept at the point (2,0,0).			1	3	7
	b.	State Gauss's law. Deduce the expression for Electric field due to an infinite line charge.			1	3	7
OR							
2	a.	Define conduction current density. Derive the expression for continuity equation.			1	3	7
	b.	The dielectric medium1 has permittivity $\epsilon_{r1} = 2$, dielectric medium2 has permittivity $\epsilon_{r2} = 4$ and the electric field in medium1 is $\mathbf{E}_1=5\mathbf{a}_x + 3\mathbf{a}_y - 13/z\mathbf{a}_z$ v/m. Determine electric field in medium2 $\mathbf{E}_2$.			1	3	7
UNIT-II							
3	a.	Explain about Biot Savart's law in detail			1	3	7
	b.	The Magnetic field intensity in a region of space is $\mathbf{H}=(x+2y)/z^2\mathbf{a}_y + 2/z\mathbf{a}_z$ A/M. Determine $\mathbf{J}$ and also the total current passing through the surface $Z=4, 1\leq X\leq 2, 3\leq Y\leq 5$.			1	3	7
OR							
4	a.	Determine the magnetic potential at a distant point due to a small circular loop of radius a carrying a current I amperes.			1	3	7
	b.	Explain about the Force due to the magnetic field on a moving charge and on a current element.			1	3	7
UNIT-III							
5	a.	Deduce the expressions for Maxwell's equations in both differential and integral form.			2	3	7
	b.	Given $\mathbf{E}=10\sin(\omega t-\beta z)\mathbf{a}_y$ V/m in free space, determine $\mathbf{D}$, $\mathbf{B}$ and $\mathbf{H}$.			2	3	7
OR							
6	a.	Deduce the boundary conditions for $\mathbf{E}$, $\mathbf{H}$, $\mathbf{D}$ and $\mathbf{B}$ at the interface between two different dielectric media.			2	3	7
	b.	A medium is characterized by $\sigma = 0$, $\mu = 2\mu_0$, and $\epsilon = 5\epsilon_0$. If $\mathbf{H} = 2\cos\{(\omega t - 3y)\}\mathbf{a}_z$, A/m, calculate ω and $\mathbf{E}$.			2	3	7
UNIT-IV							

7	a.	Define uniform plane wave. Deduce the expression for intrinsic impedance of free space.	3	3	7
	b.	State wave propagation characteristics of good conductors. A medium is defined by $\sigma = 5.8 \times 10^7$ mho/m, $\epsilon_r = 1$, $\mu_r = 1$, supports a uniform plane wave of frequency 60Hz. Determine the attenuation constant, propagation constant, intrinsic impedance, wavelength and phase velocity.	3	3	7
OR					
8	a.	List different types of polarization. Explain about circular polarization of a uniform plane wave.	3	3	7
	b.	Derive the expression for reflection and transmission coefficients, when a uniform plane wave incident normally on the surface of a perfect dielectric.	3	3	7
UNIT-V					
9	a.	A distortionless line has $Z_0 = 50\Omega$, $\alpha = 20$ m Np/m, $V = 0.6C$, where C is the speed of light in vacuum. Compute R, L, G, C and λ at 100 MHz.	4	3	7
9	b.	Derive the input impedance of a Transmission line in terms of characteristic impedance.	4	3	7
OR					
10	a.	A rectangular waveguide has cross-sectional area of $2.29 \times 1.45 \text{ cm}^2$ and operating frequency is 10 GHz. Calculate (a) free space wavelength (b) cut-off wavelength (c) cut-off frequency (d) angle of incidence (e) guide wavelength (f) phase velocity (g) phase shift constant and wave impedances of the waveguide (TE and TM).	4	3	7
	b.	Explain the reason for inability of rectangular waveguide to allow TEM mode propagation.	4	3	7